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Mechanisms of cerebellar gait ataxia

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Abstract
The cerebellum is important for movement control and plays a critical role in balance and locomotion. As such, one of the most characteristic and sensitive signs of cerebellar damage is gait ataxia. How the cerebellum normally contributes to locomotor behavior is unknown, though recent work suggests that it helps generate appropriate patterns of limb movements, dynamically regulate upright posture and balance, and adjust the feedforward control of locomotor output through error-feedback learning. The purpose of this review is to examine mechanisms of cerebellar control of locomotion, emphasizing studies of humans and other animals. Implications for rehabilitation are also considered.

Key words: *Ataxia, coordination, dysmetria, gait, motor, movement disorder, multijoint, stroke, tremor*

Introduction

The most striking and debilitating effect of damage to the cerebellum is ataxia, a word that comes from the Greek and translates literally to "without order". Ataxia generally refers to incoordination of movement. Often it implies incoordination of gait. Cerebellar gait has been described as "drunken" gait because individuals with this disorder tend to stumble and veer, resembling a person who is intoxicated. A careful examination and review of the anatomical, physiological and behavioral evidence suggests that the cerebellum may play a role in many aspects of gait, with different regions of the cerebellum responsible for different features of locomotor control.

Clinical signs of cerebellar gait ataxia

Babinski and Holmes were two of the earliest to describe many of the signs now considered classic features of gait ataxia. Their early works showed that cerebellar patients have severe difficulty maintaining balance during walking, leading to falls. For many patients, the balance deficits were found to be similar with or without the use of visual feedback (1,2). Holmes reported specifically that walking is slowed, with steps that are short, irregular and unequal in length. The leg on the more affected side is sometimes lifted overly high during swing phase by excessive flexion at the hip and knee, and then lowered abruptly and with uncontrolled force. The trajectory of walking often veers erratically and patients have difficulty with stops or turns, especially if performed quickly (2).

Much was learned from these early observations and many studies since have attempted to more specifically quantify the deficits associated with cerebellar gait ataxia. It is now well established that patients with cerebellar damage walk without the consistency in timing, length and direction of steps typical of healthy adults (3,4). In some cases gait appears wide-based. There is also increased variability in both the timing and excursion at the hip, knee and ankle joints, and the trajectory of the foot during swing is irregular. Interjoint coordination patterns across the hip, knee and ankle are abnormal (3–5). The abnormal patterns include either an irregular timing of peak flexion at one joint with respect to the other joints and/or joint-joint decomposition, a reduction in the movement at one joint during movement at another joint, in order to convert a multijoint movement into a series of single-joint movements.

Cerebellar functional localization

The arrangement of cells in cerebellar cortex is highly invariant across its entirety. However, the

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cerebellum can be divided into distinct functional zones based on its afferent and efferent connectivity (6–8). The modern view is that all regions of the cerebellum probably play an important role in control and adjustment of balance and locomotion, though each in different ways. The most medial zone of the cerebellum, the vermis, receives input from the primary vestibular afferents, vestibular, reticular and pontine nuclei, and from the spinal cord (9–13). The dorsal spinocerebellar tract (DSCT) conveys information about the sensory state of the limbs (14), whereas the ventral spinocerebellar tract (VSCT) also carries information about motor commands from spinal interneurons (14–16). The medial zone projects out via fastigial nuclei primarily to vestibular and reticular nuclei (17,18). Thus, the medial cerebellar zone can integrate spinal and vestibular inputs and influence important motor pathways for walking (i.e., vestibulospinal and reticulospinal tracts). Far less dense outputs also exist from fastigial nuclei to the motor cortex (via the thalamus) that may provide an additional route for the midline cerebellum to modulate locomotor behavior (17,19). The intermediate zone of the cerebellar hemisphere receives inputs from the DSCT and VSCT, reticular nuclei and cerebral cortical areas (10–13,20,21). This zone outputs via interpositus nuclei (or globose and emboliform nuclei in man) to the red nucleus and the cerebral cortex via the thalamus (17,19). Thus, the intermediate zone can integrate spinal and cortical inputs and influence walking through projections to motor cortical areas. The lateral zone of the cerebellar hemisphere receives input primarily from cerebral cortical areas through pontine nuclei, with the densest projections from the primary motor, premotor, primary somatosensory, and posterior parietal cortices, and also some from prefrontal and temporal cortical areas (20–22). The lateral zone projects out via dentate nuclei to the red nucleus and to many cortical regions via the thalamus, including primary motor, premotor, parietal, and prefrontal regions (18,23–28). Thus, the lateral cerebellum may influence walking via cortical interactions and may be most important for voluntary modifications to the locomotor cycle. The flocculonodular lobe can be considered a separate functional region. This lobe receives inputs from the primary vestibular afferents, vestibular nuclei, and reticular nuclei (9–11); the paraflocculus receives similar vestibular input and also visual inputs via pontine nuclei (29). Purkinje cells in the flocculonodular lobe project out to directly influence the vestibular nuclei for the control of eye movements, balance and locomotion (30–31).

Influences of the flocculonodular lobe and the medial zone of the cerebellum on balance and locomotion are well documented in animals. Lesions of the vestibular or fastigial nuclei in cats and monkeys cause abnormal changes in upright postural tone (32–34), impairments in maintaining sitting and standing balance (35), and difficulty walking (32–35). Extracellular neural recordings are in agreement with the lesion data. In a series of studies, Orlovsky recorded from cells in the vestibular (36,37) and reticular (37) nuclei in decerebrate cats at rest and during brainstem-microstimulated treadmill locomotion. Both cell types were found to have increased tonic firing rates during locomotion compared to rest, and both showed frequency modulation timed to particular phases of the step cycle. Vestibular neurons were usually more active during stance phase, with increased firing rates associated with increased extensor muscle activity. Reticular neurons were usually more active during swing phase, with increased firing rates associated with increased flexor muscle activity (37). Cerebellar influence over the vestibular nuclei was demonstrated by recording from these neurons before and after removal of the entire cerebellum (36): After cerebellectomy, there was an overall increase in firing rates in the vestibular nuclei and widespread loss of their modulatory characteristics (Figure 1A).

The role of the intermediate region of the cerebellum in locomotion is not as well understood. Several studies have reported that walking deficits are less pronounced after damage to intermediate cerebellar regions compared to more medial regions (33–35). Upright posture and balance show little or no impairment during standing and walking in cats and monkeys with intermediate zone cerebellar damage (33–35). Cats with intermediate anterior lobe or interpositus lesions can often walk rather effectively overground, but have difficulty walking on a treadmill (38) or on rungs of a ladder (34). Some notable abnormalities of walking that have been reported include a loss of placing reflexes (33), ipsilateral limb hypermetria during swing phase (33,34,38), and abnormal timing of forelimb stance and swing durations (39). With respect to this latter impairment, lesions of the intermediate cerebellar cortex have been shown to result in an increased swing phase and a decreased stance phase in the ipsilateral limb, and a decreased swing phase and an increased stance phase in the contralateral limb (39). This pattern of movement is associated with enhanced ipsilateral flexor muscle activity in late stance and throughout swing phase (40). Recordings of simple spikes from Purkinje cells in the cat paravermal cortex (41,42) or from the interpositus nucleus (41) show step cycle related rhythmic modulation during locomotion. For most cells, the greatest activity from Purkinje cells occurs during the transition between ipsilateral stance and swing phases and the least activity occurs during mid-stance (42).

Typically, animal lesion studies show that overground walking deficits are minimal after lateral

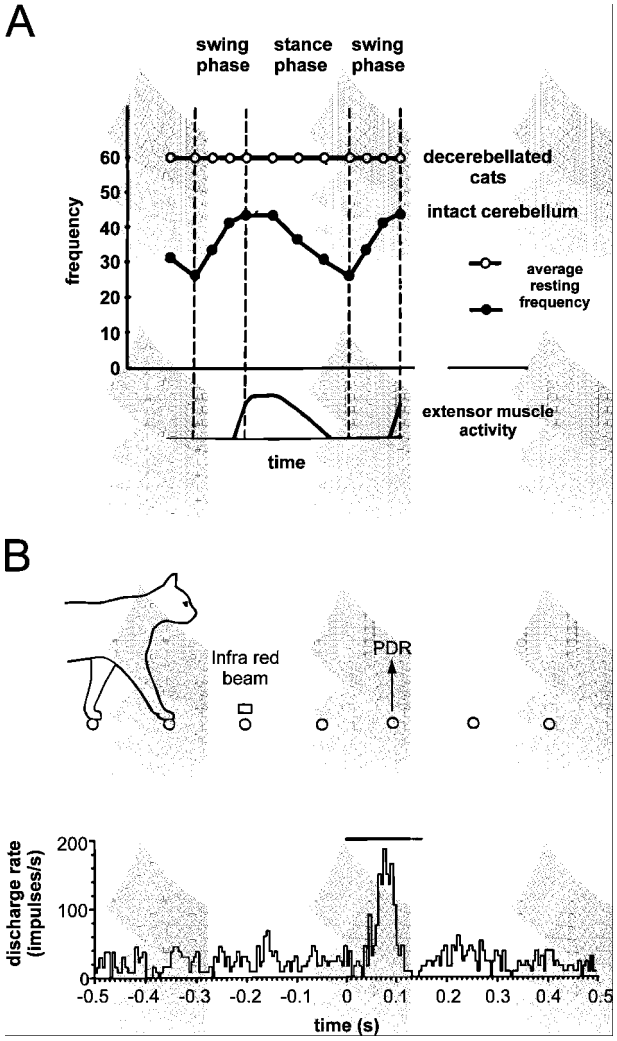

Figure 1. Cerebellar activity and walking. (A) Vestibular neurons that are normally active during periods of extensor muscle activity lose their modulatory characteristics after cerebellectomy. Top two traces show discharge rates from a typical vestibular neuron during locomotion in decerebellate (open circles) and intact (filled circles) cat. Bottom trace shows a schematic of extensor muscle activity from the ipsilateral hindlimb during locomotion. Average discharge frequencies during rest are shown at right. Adapted with permission from Elsevier. (B) Dentate neurons can show frequency modulation in response to a behaviorally relevant stimulus during walking. A pre-displaceable rung (PDR) was made to move upward when triggered by cats' limbs crossing in front of an infrared (IR) beam aimed at the walkway. An example dentate neuron fired in response to rung movement during locomotion (averaged over 27 trials, bin width=5 ms). Horizontal bar indicates the duration of rung movement (starting at time zero). Importantly, the response to rung movement during walking is greater than that seen in response to a visual stimulus alone or to rung movement that is irrelevant to the ongoing motor behavior. Adapted with permission from the Physiological Society.

cerebellar damage (34,35). The most consistent finding in the literature is an obvious misplacement of the ipsilateral limbs during difficult locomotor tasks, such as beam walking (34). Recording studies in decerebrate cats have shown that cells in the dentate nucleus modulate their activity during specific periods of a locomotor perturbation, e.g., when the treadmill is suddenly stopped (43). Purkinje cells in the lateral cortex and cells in the dentate nuclei of cats also show step cycle related rhythmical modulation during locomotion on horizontal rungs of a circular ladder (44). Some of these neurons also show frequency modulation according to unexpected movements of ladder rungs that are located just ahead of the walking cats. When a ladder rung located in front of the walking cat is elevated, some dentate neurons increase their firing rates (45). This is presumed to reflect a response to a behaviorally relevant visual cue, rather than a simple response to a visual stimulus, because the magnitude of neural responses is scaled to the immediacy of the required motor response and that neural responses become markedly shorter and of much smaller amplitude when the cat is presented with visual cues that are irrelevant to the ongoing motor behavior (Figure 1B).

In summary, the medial cerebellar region plays a primary role in regulating extensor tone, sustaining upright stance and dynamic balance control, and modulating the rhythmic flexor and extensor muscle activity that makes up the locomotor pattern. In contrast, the intermediate cerebellar region plays a fairly minimal role in controlling stance and posture. Instead, it appears to be more important for directing limb placement and regulating agonist-antagonist muscle pairs to control the relative timing, amplitude, and trajectory of limb movements, especially in circumstances when more precision is required. The lateral cerebellum is less important for the control of balance and uninterrupted level walking. However, it does seem to play a significant role in making adjustments to the normal locomotor pattern in novel or complex circumstances or when strong visual guidance is required.

Mechanisms of cerebellar gait ataxia in humans

Human studies of cerebellar involvement in balance and locomotion are more limited than the extensive animal literature. Cerebellar damage and disease occurs somewhat infrequently and is often accompanied by injury to other central nervous system regions like the brainstem or spinal cord. This makes determination of the specific role of the cerebellum in balance and locomotor control more difficult. Additionally, even focal cerebellar lesions in humans often span multiple longitudinal zones (46), making it hard to localize functions to a particular cerebellar region.

Postural instability

Cerebellar damage in humans typically results in increased standing postural sway (47–49). Notably, cerebellar deficits of sway (measured by center-of-pressure movement recordings on a forceplate) are

specific and depend upon the location of the lesion (47,50). It is generally found that cerebellar anterior lobe damage produces balance abnormalities characterized by increased postural sway of high velocity and low amplitude in the anterior-posterior dimension, postural tremor, and increased intersegmental movements of the head, trunk and legs. In contrast, vestibulocerebellar lesions lead to increased postural sway of low frequency and high amplitude and without a preferred direction or any increased intersegmental movements. Lateral cerebellar lesions tend to produce only slight postural instability that is almost indistinguishable from healthy individuals (47,48,50) (Figure 2).

By perturbing the support surface, more dynamic postural responses can be examined. Using this technique, it has been shown that cerebellar damage is associated with hypermetric postural responses to backward surface displacements (51). Specifically, individuals with cerebellar damage show hypermetric postural responses and overshooting of the initial posture, with larger than normal surface-reactive torque responses and exaggerated and prolonged muscle activity. More recently, it was shown that patients with degenerative cerebellar damage have impaired dynamic postural responses in the lateral as well as the backward direction. The impaired responses seem to be associated with both excessive (hypermetric) movements of the trunk, but also reduced movements, or "stiffening", of the knees (52). During gait, trunk sway is also increased, especially in the anterior-posterior dimension (53).

The extent to which balance abnormalities contribute to cerebellar gait ataxia has recently been examined (54). In a group of individuals with varying levels of balance and voluntary leg coordination deficits, it was found that the subjects with balance deficits (and no leg coordination deficits) showed many of the characteristic features of cerebellar gait ataxia, including reduced stride lengths, increased stride widths, reduced joint excursions, abnormal swing foot trajectories, increased variability in foot placement, decomposition between the ankle and knee joints, and reduced walking speed. The subjects with leg coordination deficits (and no balance deficits) did not have significant walking impairments (54). Therefore, imbalance appears to contribute much more strongly to cerebellar gait ataxia than does leg incoordination, at least during uninterrupted level walking. Interestingly, in a group of children who underwent surgical split of the cerebellar vermis for removal of tumors of the fourth ventricle, overground walking and hopping on one foot were found to be rather well-preserved, while tandem walking (placing one foot directly in front of the other) was disrupted (55). This surprising result may indicate that the vermis is not only critical for posture and balance, but also for precise interlimb coordination during locomotion (perhaps via cerebellar parallel fibers crossing the midline).

Decomposition of multijoint leg movements

Decomposition of movement is commonly seen in individuals with cerebellar gait ataxia (3–5). Decomposition refers to the strategy of breaking down a multijoint movement, such as combined hip, knee, and ankle rotations, into a series of single joint movements (2,56), presumably to compensate for impaired multijoint coordination. Earhart and Bastian quantified the amount of leg decomposition in a group of control subjects and a group of individuals with cerebellar damage walking over level and inclined surfaces (Figure 3). Overall, cerebellar damage led to increased joint-joint decomposition that tended to worsen as the surface inclination increased (5).

Decomposition of leg movements during gait may be an attempt to compensate for poor control of intersegmental dynamics, though this issue has not yet been addressed in the literature. Several studies have shown that cerebellar damage causes specific deficits in the control of interaction torques, those torques produced at a segment that are caused by movements at other linked segments, during

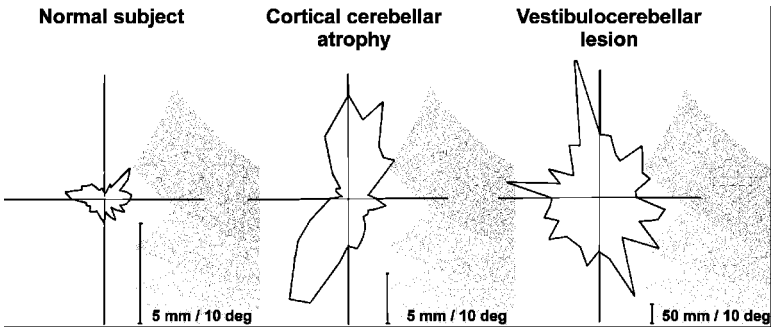

Figure 2. Sway patterns associated with different cerebellar lesions. Sway position (center of pressure) histograms (bin width 10° polar; sampling frequency 30 ms; time 60 s) from a typical control subject (left), a cerebellar subject with anterior lobe cortical atrophy (middle), and a cerebellar subject with a vestibulocerebellar medulloblastoma (right). Note the scale changes across plots. The cerebellar subject with anterior lobe atrophy shows moderate instability in the anterior-posterior dimension. The cerebellar subject with vestibulocerebellar damage shows severe instability that is omnidirectional. Adapted with permission from Lippincott, Williams and Wilkins.

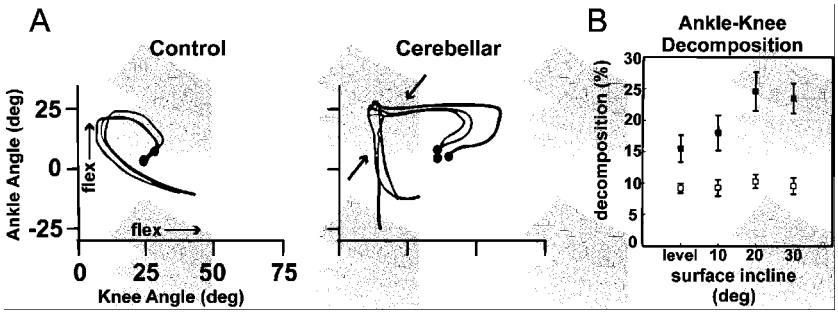

Figure 3. Joint movements are decomposed during walking. (A) Angle-angle plots depict ankle angle relative to knee angle movements from several trials from a typical control subject (left) and a cerebellar subject with an infarct of the superior cerebellar artery (right) during stance phase of walking over a 30° incline. Circles show stance phase onset. Arrows on the cerebellar plot show the decomposition (e.g., when one joint moves while the other is kept fixed), indicated by the horizontal and vertical components of the trace. (B) Average (± 1 SE) ankle-knee decomposition for control (open squares) versus cerebellar (filled squares) groups during stance phase walking over level and inclined surfaces. Decomposition is quantified as the percentage of time in stance that the ankle or knee joint is kept still (angular velocity $< 5°/s$) while the other joint is moving (angular velocity $\geq 5°/s$). Cerebellar subjects show increased decomposition overall compared to controls, with worsening effects as the surface incline rises. Adapted with permission from the American Physiological Society.

reaching movements (57–60). Hypermetria, or overshooting of targets, and curved hand paths are hallmarks of cerebellar dysfunction (3,56,58,59,61) that have been shown to be related to impaired control of interaction torques during reaching (57–60). Cerebellar leg hypermetria (in the form of excessive foot elevation) during stepping in place over a small obstacle seems to be related to an overcompensation for interaction torques at the knee (62), but it is not known whether specific features of cerebellar gait ataxia may also be related to poor control of intersegmental dynamics, e.g., abnormal and irregular foot placement, excessive foot path curvature, impaired interjoint coordination of the stance limb.

Reduced adaptive capabilities for posture and locomotion

It is well known that cerebellar damage impairs motor adaptations of several types of specific arm (63–65) and eye movements (66–69), though relatively few studies have addressed adjustments of locomotor patterns.

In one study, adjustments in torque responses were measured during standing postural perturbations in the form of a backward translation of the support surface. When perturbations were presented in a blocked (predictable) order, healthy subjects, but not individuals with cerebellar damage, were able to appropriately scale their anticipatory postural responses (51). This suggests that the cerebellum plays an integral role in using predictive feedforward control to adapt postural responses. Another study reported a similar deficit during a locomotor-like adaptation paradigm (70). Following repeated stepping in place (with the body stationary in space) on a rotating disc, healthy individuals demonstrate a 'podokinetic after-rotation' of stepping in a curved trajectory (71), presumably caused by a recalibration of proprioceptive information between the trunk and stance limbs. Cerebellar damage interferes with the ability to store this novel body orientation during locomotion, however, indicated by reduced amplitudes of the after-rotation (70).

Recently, effects of cerebellar damage on locomotor adaptation to a splitbelt treadmill have been reported (72). Locomotion on a splitbelt treadmill, when one leg is forced to move faster than the other, has been shown to produce both reactive and predictive types of motor adaptations (73,74). The reactive adaptations arise quickly following the splitbelt perturbation and similarly disappear quickly upon removal of the splitbelt condition (Figure 4A, 4B). The predictive adaptations require several strides to become apparent and also show negative aftereffects upon return to the non-perturbed state (Figure 4C, 4D). The presence of aftereffects (persistence of the new motor behavior) suggests that the nervous system has stored some critical information about the state of the body or its environment. Cerebellar damage results in significant impairments of the slower-forming, predictive adaptations, specifically, slower adaptations and reduced aftereffects, but does not interfere with the faster-forming, reactive adaptations (72).

Overall, growing evidence seems to support a role for the cerebellum in adapting posture and locomotion through trial-and-error practice. It seems to be particularly important for the formation or utilization of predictive control for these types of movements.

Conclusion

Grillner and Wallen (75) proposed three primary requirements of the central nervous system to produce normal locomotion. It must generate the basic pattern of rhythmic reciprocal flexor and extensor muscle activity, maintain control of equilibrium during forward movement of the center of mass, and have some capability for adapting to novel contexts or changes in behavioral goals. We propose

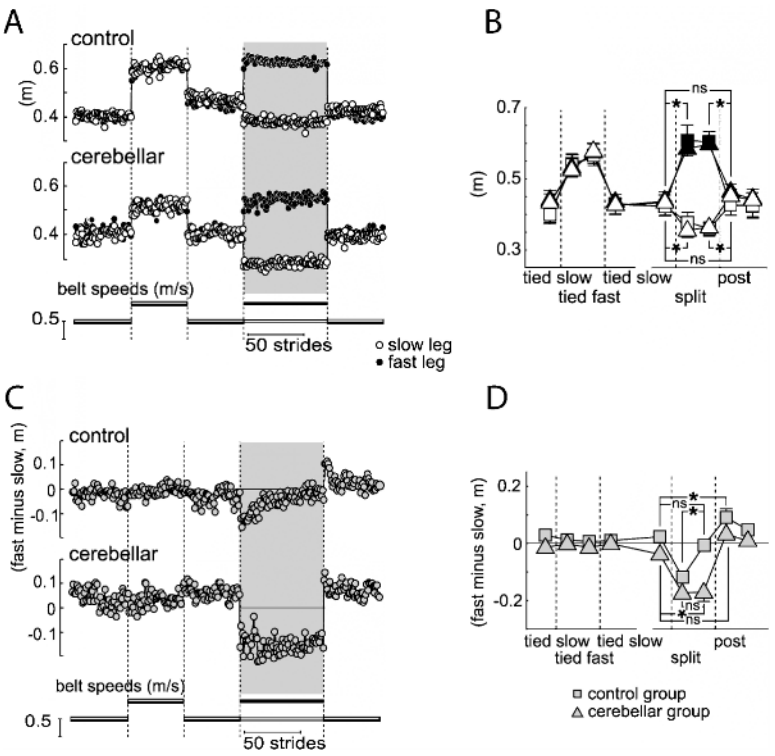

Figure 4. Cerebellar damage impairs predictive, not reactive, walking adaptation. (A–B) During splitbelt walking, both controls and cerebellar subjects can immediately react by lengthening their fast stride and shortening their slow stride. A stride is the distance traveled by one leg from initial contact to lift off of the treadmill. (A) Trial-by-trial plot of sequential strides from a typical control and cerebellar subject at baseline (both legs walking either fast or slow), splitbelt adaptation (one leg fast, one slow), and post-adaptation (both legs slow). (B) Group averages of stride lengths for the control (squares) and cerebellar (triangle) groups. Each data point represents values averaged over the early or late portions of each testing period. (C–D) Cerebellar subjects are impaired at predictive adaptation of step length (i.e. using prior errors to update subsequent steps). Note the distinction between a step and a stride: a step is the distance from one foot contact to the other foot contact, and two steps (normally symmetric) make up one stride. (C) Step length differences for sequential strides on the treadmill from a typical control and cerebellar subject across all testing periods. The grey shaded area represents the time when belts were split. Note that controls adapt during splitbelt walking to re-establish symmetry (i.e., a difference of 0). (D) Average step length differences for control (squares) and cerebellar (triangles) groups. For all plots, error bars indicate ± 1 SE. Asterisks indicate a significance level of $p<0.05$ for the post hoc analysis; ns=not significant. Adapted with permission from the *Journal of Neuroscience*.

that the cerebellum contributes to each of these requirements in specific ways.

First, while the basic locomotor synergy is likely to be generated in the brainstem and spinal cord (76,77), the cerebellum probably contributes to modulations of the timing, rate, and force of muscle activity controlling locomotor patterns (36,40,42, 43,78,79). Thus, the cerebellum helps coordinate relative motions at various joint segments both within (3–5,38) and between limbs (39,55,72,80), in order to produce the highly stereotyped walking pattern. The medial and intermediate zones of the cerebellum would play a critical role in this function.

Second, the cerebellum is necessary for control of upright posture during walking. Cerebellar damage causes impaired sitting, standing, and walking balance, abnormal postural tone, and frequent falling during locomotion (32–35,47–49). The cerebellum utilizes both anticipatory (81) and feedback (51) mechanisms to maintain balance during locomotion, and when those mechanisms fail, gait ataxia results (54). The medial cerebellar zone probably plays the strongest role here.

Finally, the cerebellum seems to be important for providing adaptability to the locomotor pattern. Patients with cerebellar damage show consistent impairments in adapting to perturbations during standing (81) or walking (82). They are particularly compromised in adaptations requiring prediction based on trial-and-error practice (70,72,83). This function may be governed by any of the distinct cerebellar regions, depending upon whether the adapted movement specifically involves brainstem postural responses (medial zone), limb placement and agonist-antagonist control (intermediate zone) or especially novel contexts or those requiring strong visual guidance (lateral zone).

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